# JAYPEE INSTITUTE OF INFORMATION TECHNOLOGY , NOIDA

## B.TECH SEMESTER -IV

## PBL REPORT

## Introduction to VLSI Lifecycle (24B25EC213)

# RTL TO GDSII OF 4x1MUX

**Submitted by**

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# Introduction :-

The process of converting a Register Transfer Level (RTL) description of a digital circuit (like a 4:1 MUX) to a GDSII layout file is a key part of VLSI physical design flow. GDSII (Graphic Data System II) is the industry-standard file format used to represent the final physical layout of an integrated circuit.

# Steps :-

## 1. RTL Design

Here is the Verilog code and testbench code we have written for our 4x1 MUX.
The process will be done on Cadence SimVision software.

**VERILOG CODE**

```
module mux4x1_case (
   input wire [3:0] in,   // 4 data inputs
   input wire [1:0] sel,  // 2-bit select signal
   output reg out         // Output of the MUX
);
   always @(*) begin
      case (sel)
         2'b00: out = in[0];  // Select input 0
         2'b01: out = in[1];  // Select input 1
         2'b10: out = in[2];  // Select input 2
         2'b11: out = in[3];  // Select input 3
         default: out = 1'b0; // Default case (if needed)
      endcase
   end
endmodule
```

**Testbench**

```
`timescale 1ns / 1ps
module mux4x1_tb;
   // Testbench signals
   reg [3:0] in;       // 4 data inputs
   reg [1:0] sel;      // 2-bit select signal
   wire out;           // MUX output

   // Instantiate the module under test (MUT)
   mux4x1_case uut (   // Replace with  mux4x1_if  if testing the  if  implementation
      .in(in),
      .sel(sel),
      .out(out) );
```

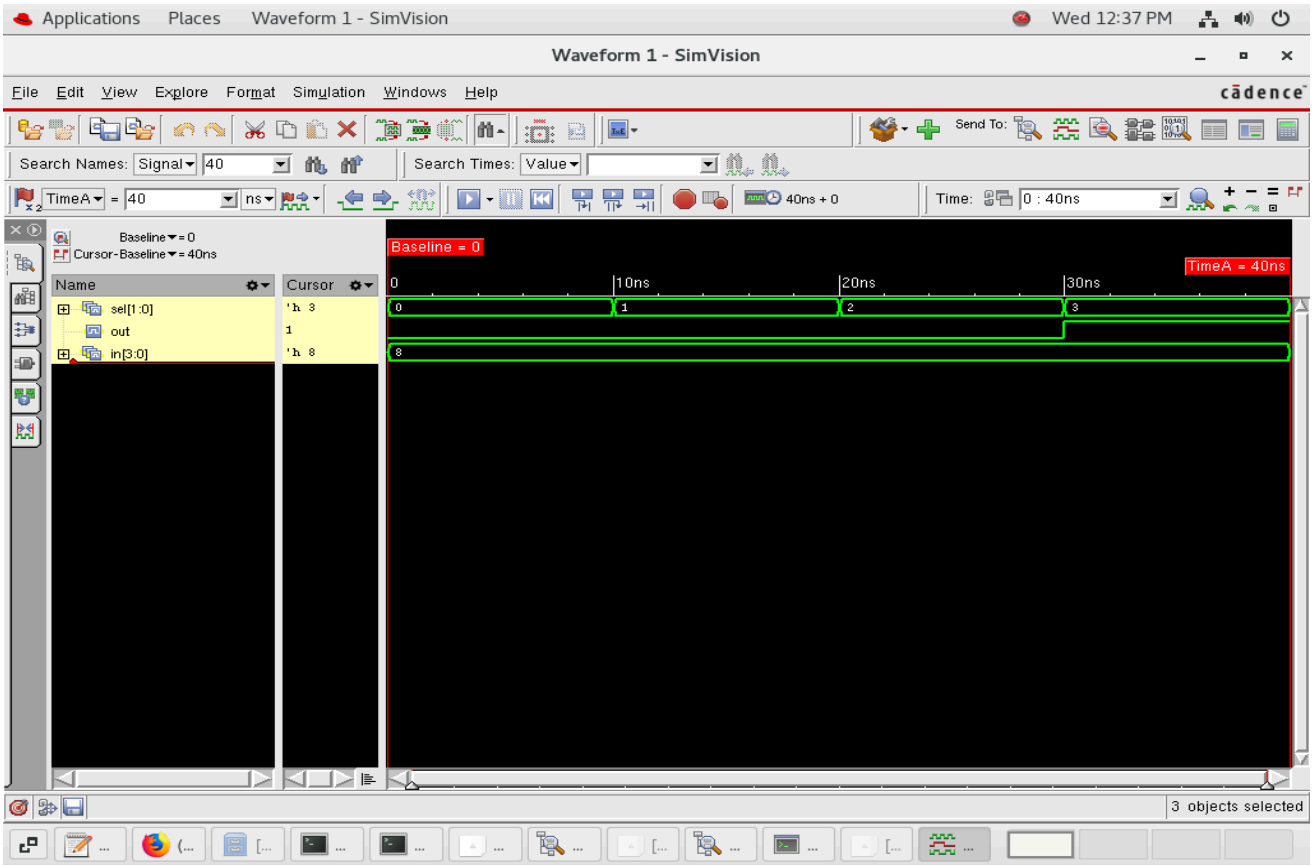
```
    // Testbench logic
    initial begin
        // Display header
        $display("Time\t sel\t in\t out");
        $monitor("%g\t %b\t %b\t %b", $time, sel, in, out);

        // Apply test vectors
        in = 4'b1101; sel = 2'b00; #10; // Select input 0
        in = 4'b1101; sel = 2'b01; #10; // Select input 1
        in = 4'b1101; sel = 2'b10; #10; // Select input 2
        in = 4'b1101; sel = 2'b11; #10; // Select input 3

        in = 4'b1010; sel = 2'b00; #10; // Test with new input vector
        in = 4'b1010; sel = 2'b01; #10;
        in = 4'b1010; sel = 2'b10; #10;
        in = 4'b1010; sel = 2'b11; #10;

        // Finish simulation
        $stop;
    end
endmodule
```

**Waveform Output**

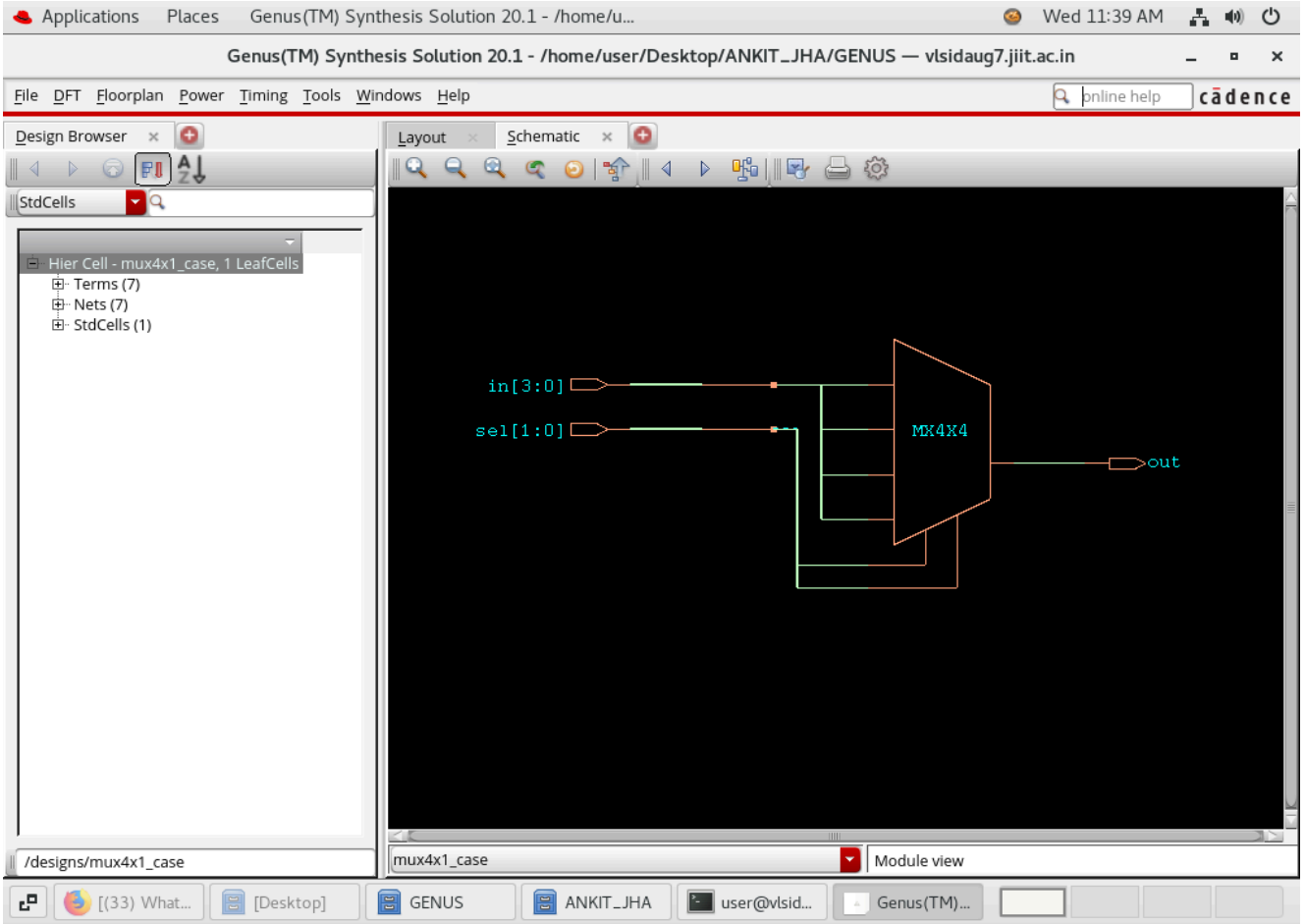


## 2. Synthesis

Converts RTL into a gate-level netlist using a standard cell library and optimizes for area, power, and timing.

The process will be done on Cadence Genus software.

**OUTPUT**



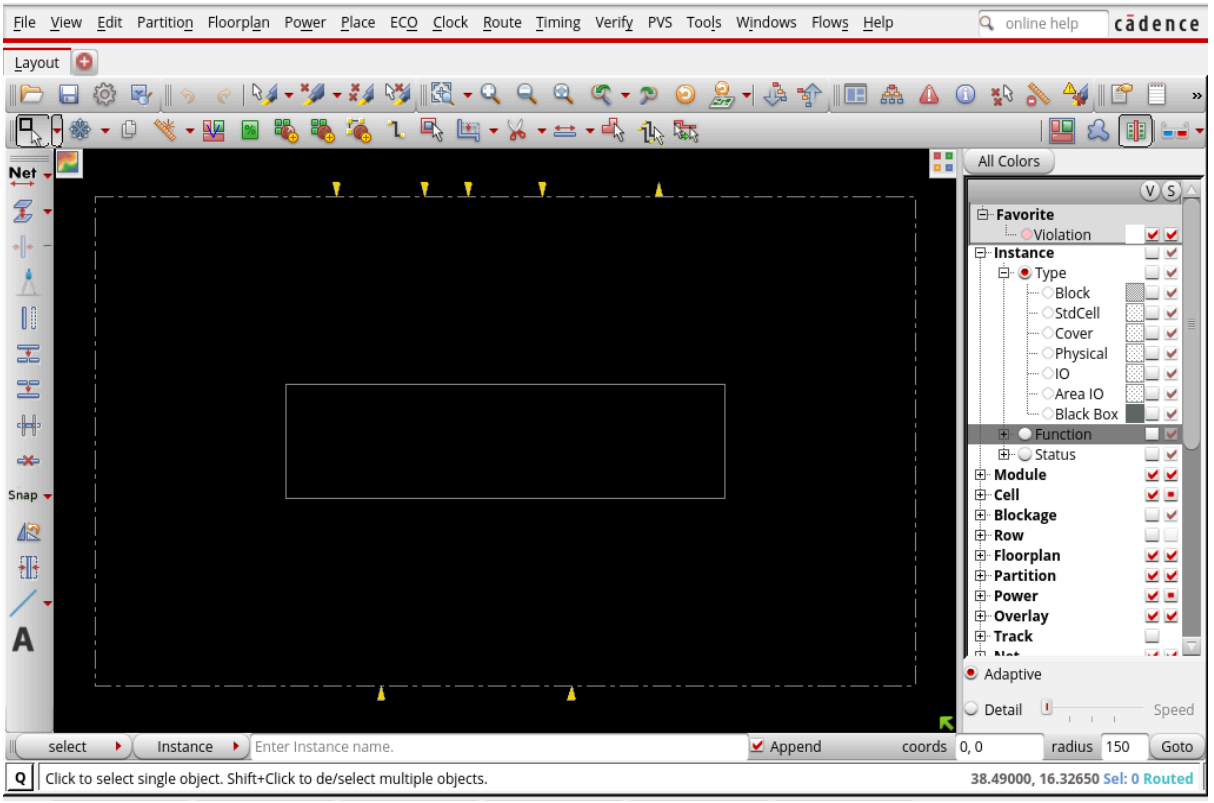

## 3. Planning, Placement and Routing

**Planning:-** Defines chip outline, IO pin placement, floor and power planning.

**Placement:-** Cells from the netlist are placed on the chip.

**Routing:-** Connects all placed cells using metal layers.

The process will be done on Cadence Innovus software.


File View Edit Partition Floorplan Power Place ECO Clock Route Timing Verify PVS Tools Windows Flows Help
online help
cadence
Layout
Net
Snap
A
All Colors
Favorite
Violation
Instance
Type
Block
StdCell
Cover
Physical
IO
Area IO
Black Box
Function
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Cell
Blockage
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Floorplan
Partition
Power
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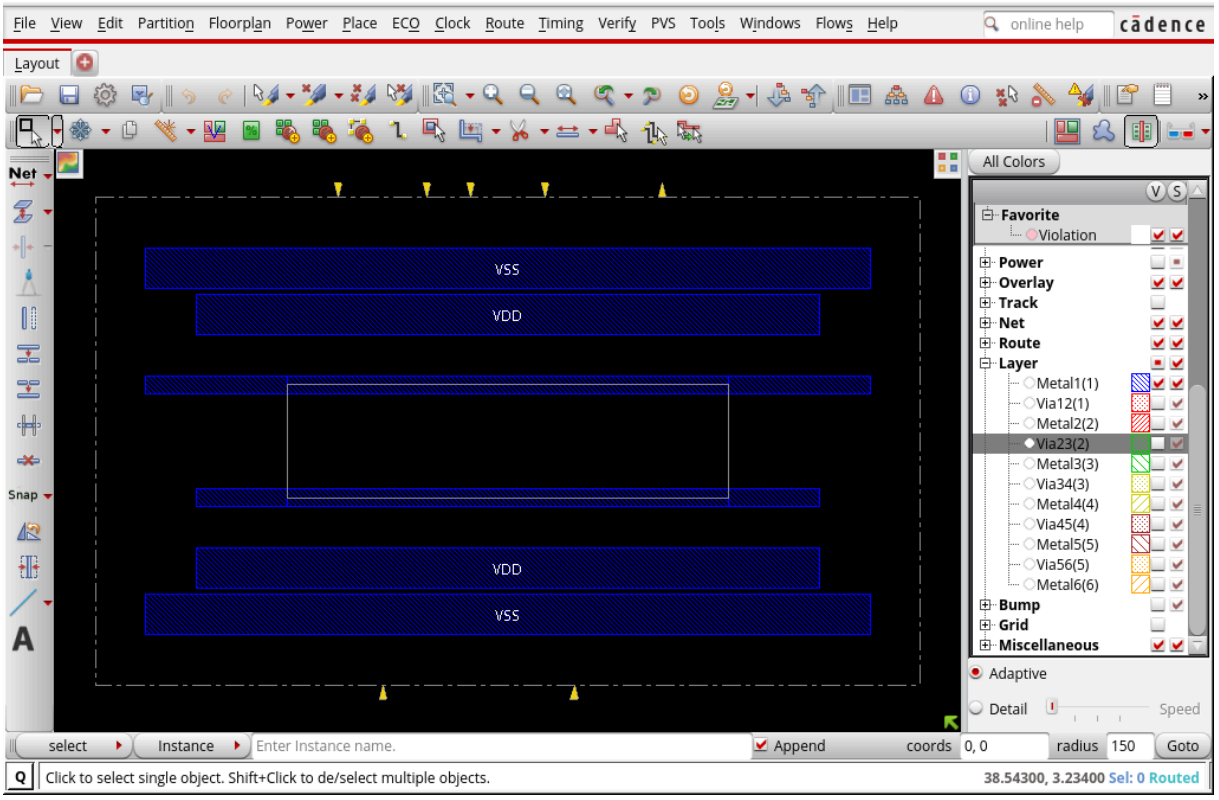
Track
Adaptive
Detail
Speed
select
Instance
Enter Instance name.
Append
coords
0, 0
radius
150
Goto
Click to select single object. Shift+Click to de/select multiple objects.
38.49000, 16.32650 Sel: 0 Routed


File View Edit Partition Floorplan Power Place ECO Clock Route Timing Verify PVS Tools Windows Flows Help
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VSS
All Colors
Favorite
Violation
Power
Overlay
Track
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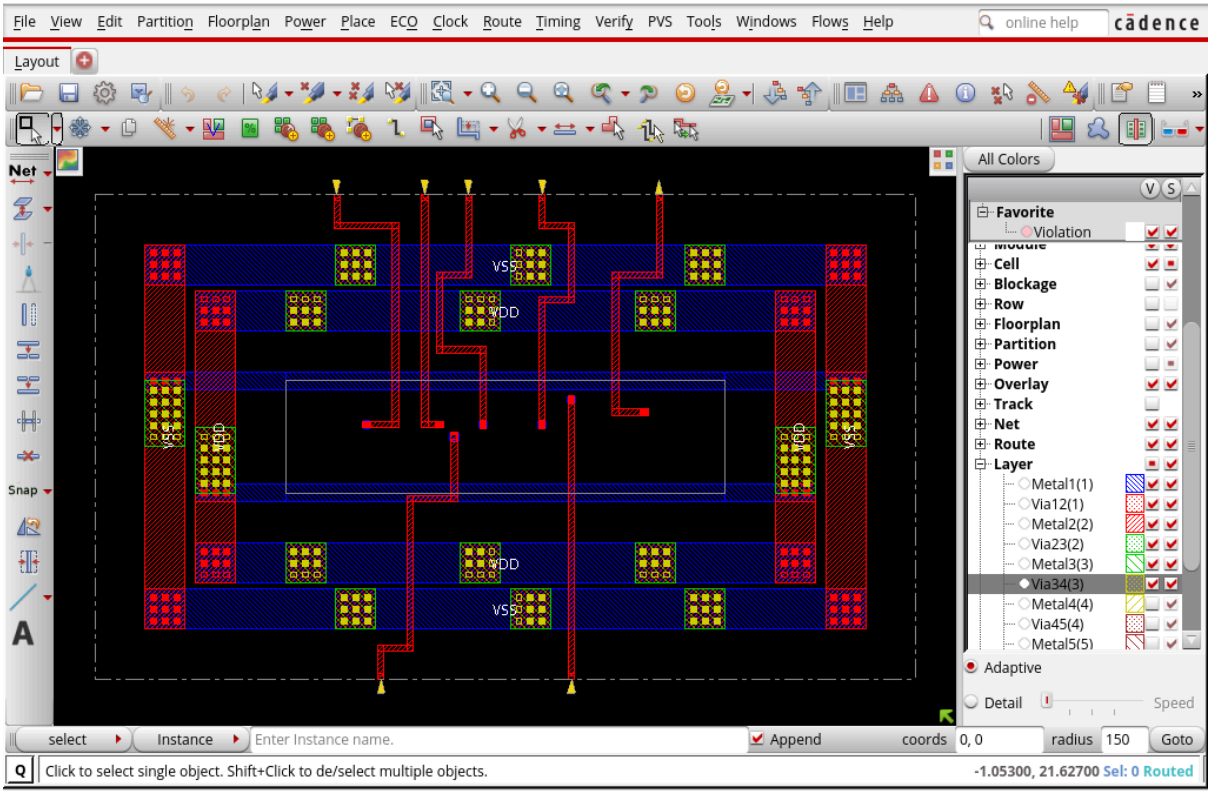
Route
Layer
Metal1(1)
Via12(1)
Metal2(2)
Via23(2)
Metal3(3)
Via34(3)
Metal4(4)
Via45(4)
Metal5(5)
Via56(5)
Metal6(6)
Bump
Grid
Miscellaneous
Adaptive
Detail
Speed
select
Instance
Enter Instance name.
Append
coords
0, 0
radius
150
Goto
Click to select single object. Shift+Click to de/select multiple objects.
38.54300, 3.23400 Sel: 0 Routed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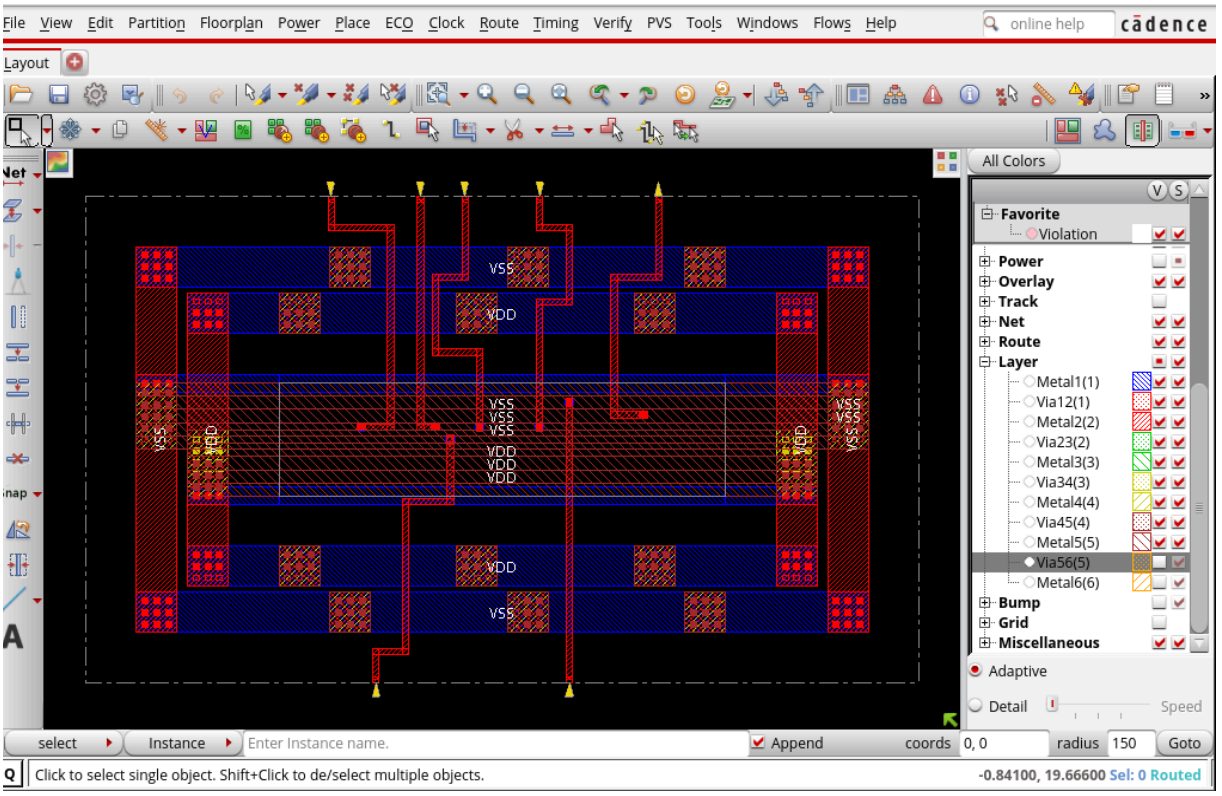
Via34(3)
Metal4(4)
Via45(4)
Metal5(5)
Adaptive
Detail
Speed
select
Instance
Enter Instance name.
Append
coords
0, 0
radius
150
Goto
Click to select single object. Shift+Click to de/select multiple objects.
-1.05300, 21.62700 Sel: 0 Routed


File View Edit Partition Floorplan Power Place ECO Clock Route Timing Verify PVS Tools Windows Flows Help
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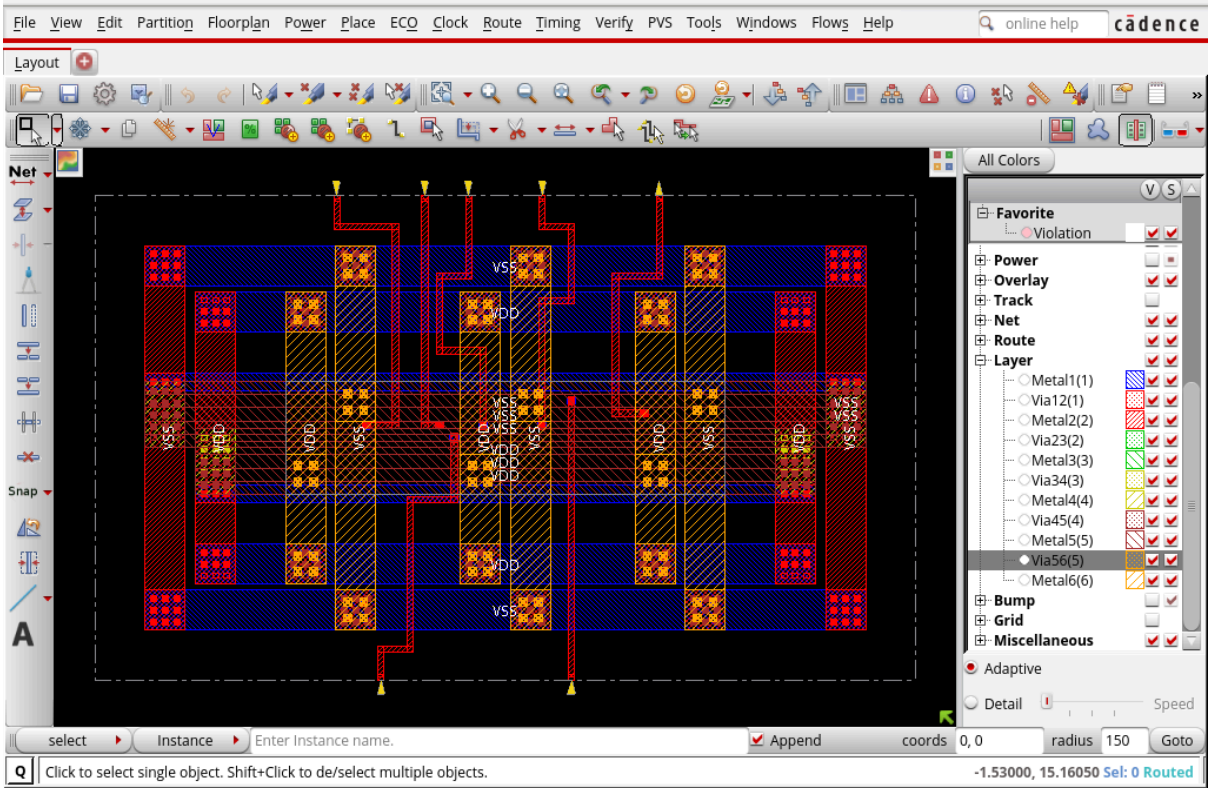
Metal1(1)
Via12(1)
Metal2(2)
Via23(2)
Metal3(3)
Via34(3)
Metal4(4)
Via45(4)
Metal5(5)
Via56(5)
Metal6(6)
Bump
Grid
Miscellaneous
Adaptive
Detail
Speed
select
Instance
Enter Instance name.
Append
coords
0, 0
radius
150
Goto
Click to select single object. Shift+Click to de/select multiple objects.
-0.84100, 19.66600 Sel: 0 Routed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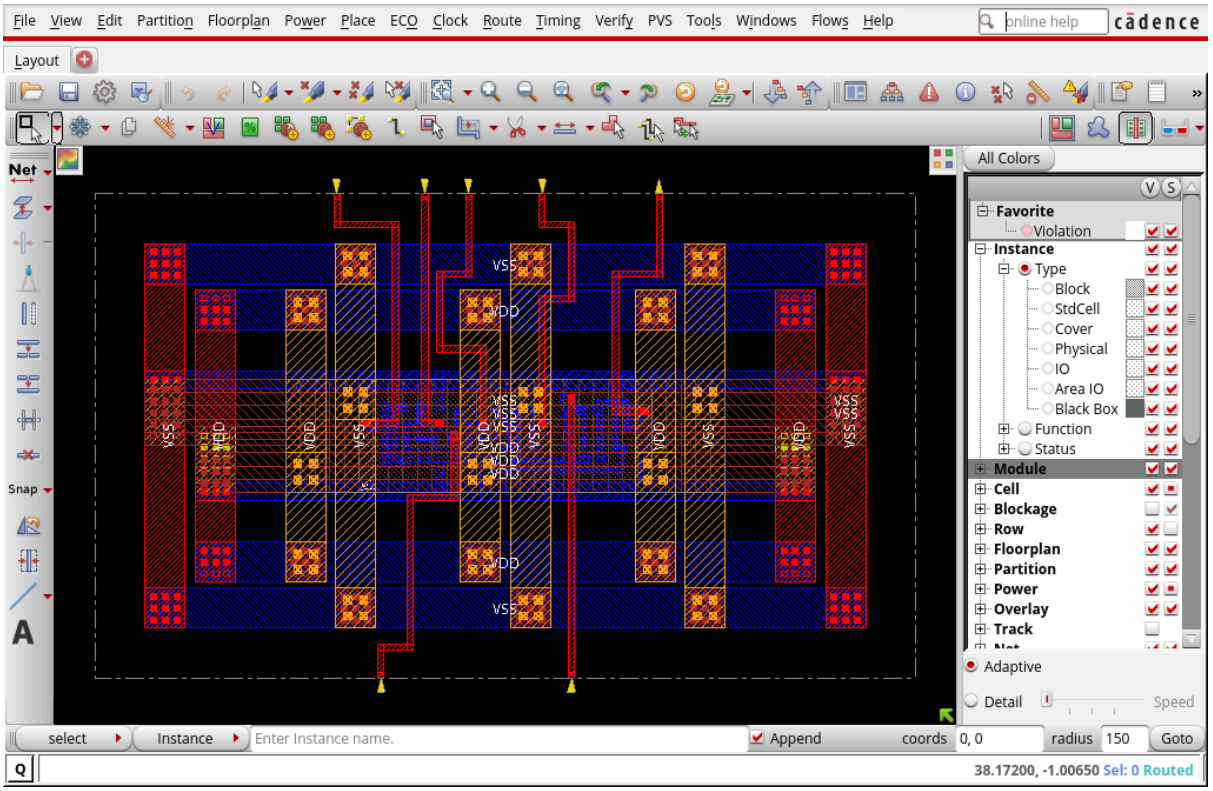
Metal2(2)
Via23(2)
Metal3(3)
Via34(3)
Metal4(4)
Via45(4)
Metal5(5)
Via56(5)
Metal6(6)
Bump
Grid
Miscellaneous
Adaptive
Detail
Speed
select
Instance
Enter Instance name.
Append
coords
0, 0
radius
150
Goto
Click to select single object. Shift+Click to de/select multiple objects.
-1.53000, 15.16050 Sel: 0 Routed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Favorite
Violation
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StdCell
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Physical
IO
Area IO
Black Box
Function
Status
Module
Cell
Blockage
Row
Floorplan
Partition
Power
Overlay
Track
Adaptive
Detail
Speed
select
Instance
Enter Instance name.
Append
coords
0, 0
radius
150
Goto
38.17200, -1.00650 Sel: 0 Routed

# 4.Reports Generation

Timing,Area and Power reports are generated .

```
File  Edit  View  Search  Terminal  Help
*** Done Building Timing Graph (cpu=0:00:00.1 real=0:00:00.0 totSessionCpu=0:02:31 mem=1391.3M)

------------------------------------------------------------------
          timeDesign Summary
------------------------------------------------------------------

Setup views included:
 WORST

+--------------------+---------+---------+
|     Setup mode     |   all   | default |
+--------------------+---------+---------+
|           WNS (ns):|  0.000  |  0.000  |
|           TNS (ns):|  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |
|          All Paths:|    0    |    0    |
+--------------------+---------+---------+

+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000    |      0 (0)       |
|   max_fanout   |      0 (0)       |     0      |      0 (0)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+

Density: 66.667%
Routing Overflow: 0.00% H and 0.00% V
------------------------------------------------------------------
Reported timing to dir timingReports
Total CPU time: 0.25 sec
Total Real time: 1.0 sec
```

```
File  Edit  View  Search  Terminal  Help
+--------------------+---------+---------+
|     Setup mode     |   all   | default |
+--------------------+---------+---------+
|           WNS (ns):|  0.000  |  0.000  |
|           TNS (ns):|  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |
|          All Paths:|    0    |    0    |
+--------------------+---------+---------+

+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000    |      0 (0)       |
|   max_fanout   |      0 (0)       |     0      |      0 (0)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+

Density: 66.667%
Routing Overflow: 0.00% H and 0.00% V
------------------------------------------------------------------
Reported timing to dir timingReports
Total CPU time: 0.25 sec
Total Real time: 1.0 sec
Total Memory Usage: 1349.535156 Mbytes
*** timeDesign #1 [finish] : cpu/real = 0:00:00.2/0:00:00.7 (0.3), totSession cpu/real = 0:02:30.6/0:30:40.6 (0.
1), mem = 1349.5M
innovus 2>
innovus 2> report_area
    Hinst Name       Module Name          Inst Count           Total Area
-----------------------------------------------------------------------
mux_c                                          1               66.528
innovus 3>
```

```
File   Edit   View   Search   Terminal   Help

Total Power
----------------------------------------------------------------------------------------
Total Internal Power:        0.00019295            92.4668%
Total Switching Power:       0.00001552             7.4383%
Total Leakage Power:         0.00000020             0.0949%
Total Power:                 0.00020867
----------------------------------------------------------------------------------------


Group                           Internal   Switching     Leakage       Total  Percentage
                                Power      Power         Power         Power  (%)
----------------------------------------------------------------------------------------
Sequential                             0           0           0           0           0
Macro                                  0           0           0           0           0
IO                                     0           0           0           0           0
Combinational                  0.0001929   1.552e-05    1.98e-07   0.0002087         100
Clock (Combinational)                  0           0           0           0           0
Clock (Sequential)                     0           0           0           0           0
----------------------------------------------------------------------------------------
Total                          0.0001929   1.552e-05    1.98e-07   0.0002087         100
----------------------------------------------------------------------------------------


Rail                  Voltage   Internal   Switching     Leakage       Total  Percentage
                                Power      Power         Power         Power  (%)
----------------------------------------------------------------------------------------
Default                   0.9  0.0001929   1.552e-05    1.98e-07   0.0002087         100


----------------------------------------------------------------------------------------
*       Power Distribution Summary:
*              Highest Average Power:                        g2 (MX4X1):        0.0002087
*              Highest Leakage Power:                        g2 (MX4X1):         1.98e-07
*                Total Cap:      1.53297e-15 F
```

## 5. Saving GDSII File

Finally, the GDSII File is created and saved. This file contains the complete information of all layers, masks, and geometries and is sent to the fabrication foundry for chip manufacturing. The file saved as Ankit_4x1mux.gds or <filename>.gds .